METRIC

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MILITARY SPECIFICATION

SWITCHES, FIBER OPTIC, SHIPBOARD, (METRIC),
GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for fiber optic switches. An optical signal is transferred from one path to another by an external force or energy applied to the switch. Excluded from this specification are switches that contain a separate optical source and optical detector. The switches must have consistent optical performance. The switches are supplied under a MIL-STD-790 reliability assurance program.

1.2 Classification. Switches shall be classified as specified in 1.2.1 through 1.2.2 (see 6.2).

1.2.1 Type. The type designation defines the actuator (see 6.6.2) and the actuation method (see 6.6.1).

Type 1 - Mechanical latching (see 6.6.5)
Type 2 - Mechanical nonlatching (see 6.6.6)
Type 3 - Electrical latching
Type 4 - Electrical nonlatching
Type 5 - Other latching
Type 6 - Other nonlatching

1.2.2 Style. The style designation defines the number of fiber inputs (M) and the number of fiber outputs (N).

Style A - 1:1 (on/off)
Style B - 1:N (selector)
Style C - 2:2 (bypass)
Style D - M:N (exchange)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-P-35 - Passivation Treatments for Corrosion-Resistant Steel.
TT-I-735 - Isopropyl Alcohol.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6021
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MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact), Shipboard Machinery, Equipment, and Systems, Requirements for.
MIL-C-5541 - Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-H-5606 - Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance.
MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4, JP-5 and JP-5/JP-8ST.
MIL-W-16878 - Wire, Electrical, Insulated, General Specification for.
MIL-F-16884 - Fuel, Naval Distillate.
MIL-L-17331 - Lubricating Oil, Steam Turbine and Gear, Moderate Service.
MIL-L-23699 - Lubricating Oil, Aircraft Turbine Engine, Synthetic Base.
MIL-S-24725/1 - Switch, Fiber Optic, Shipboard, Electrical Nonlatching; Bypass, Multimode Cable, Stand-Alone.
MIL-S-28786 - Switches, Packaging of.
MIL-F-49291/6 - Fiber, Optical, 62.5/125 Micrometers, Radiation Hardened (Metric).
MIL-C-85045 - Cable, Fiber Optic, Shipboard (Metric) General Specification for.
MIL-C-85045/14 - Cable, Fiber Optic, One Fiber, Cable Configuration Type 2 (OFCC), Loose Tube, Cable Class MM or SM (Metric).
MIL-C-85045/16 - Cable, Fiber Optic, One Fiber, Cable Configuration Type 2 (OFCC), Tight Buffered, Cable Class MM or SM (Metric).

STANDARDS

FEDERAL

FED-STD-313 - Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-790 - Product Assurance Program Requirements for Electronic and Fiber Optic Parts Specifications.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-883 - Test Methods and Procedures for Microelectronics.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.
MIL-STD-1344 - Test Methods for Electrical Connectors.
DOD-STD-1678 - Fiber Optic Test Methods and Instrumentation.
MIL-STD-45662 - Calibration systems requirements.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Bldg. 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D-1141 - Sea Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA) AND TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

* EIA/TIA-455 - Standard Test Procedures for Fiber Optic Fibers, Cables, Transducers, Connecting and Terminating Devices.
* EIA-455-13 - Visual and Mechanical Inspection of Fibers, Cables, Connectors and/or Other Fiber Optic Devices.
* EIA-455-20 - Measurement of Change in Optical Transmittance.
EIA-455-22 - Ambient Light Susceptibility of Fiber Optic Components.
EIA-455-32 - Discontinuities.
EIA-455-34 - Interconnection Device Insertion Loss Test.
EIA-455-36 - Twist Test for Fiber Optic Connecting Devices.
EIA-455-42 - Optical Crosstalk in Fiber Optic Components.
EIA-455-50 - Light Launch Conditions for Long-Length Graded-Index Optical Fiber Spectral attenuation measurements.

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EIA-455-107 · Return Loss for Fiber Optic Components.

* DOD Adopted

(Application for copies should be addressed to the Electronic Industries Association/Telecommunications Industry Association (EIA/TIA), 2001 Pennsylvania Ave, NW, Washington, DC 20006-1813.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. The switches furnished under these specifications shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award of contract (see 4.4 and 6.4).

3.2.1 Product assurance. The contractors product assurance program for assembled switches and assembled procedures shall meet the requirements of MIL-STD-790.

3.3 Materials. Materials selected for use in switches shall be of a type and quality to ensure compliance with the requirements of this specification, and shall be physically and chemically compatible for their intended lifetime which shall be not less than 20 years. They shall be of the lightest weight practical for the intended purpose and shall in no manner interfere with the optical junction transmission and the optical switching process. Combinations of materials shall be non-nutrient to fungus. Unless otherwise specified (see 3.1), combinations of materials shall be nonmagnetic. They may be dielectric or electrically conductive as applicable. Materials used in the switch construction shall not emit toxic or explosive fumes when exposed to high temperature or flame. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency.

3.3.1 Recovered materials. Unless otherwise specified herein, all material incorporated in the products covered by this specification shall be new. Products may be fabricated using raw materials produced from recovered bulk materials to the extent practicable if the intended use of the product is not jeopardized. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become part of a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of partially processed, assembled, used or rebuilt products are allowed under this specification.

3.3.2 Metals. Metals, other than current carrying parts, shall be corrosion-resistant. Unless otherwise specified (see 3.1), metals shall be nonmagnetic and all exposed corrosion-resistant steel parts shall have a passivated surface that is compatible with external coatings or platings.

3.3.3 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless finished to prevent electrolytic corrosion. Selection and protection of dissimilar metal combinations shall be in accordance with MIL-STD-889.

3.3.4 Finish. The resultant finish on all switches shall be:

a. Stainless steel components - passivate in accordance with QQ-P-35.
b. Brass - titanium coating for internal parts.

3.3.5 Material safety data sheet (MSDS). The contracting activity shall be provided a material safety data sheet at the time of contract award. The MSDS shall be provided in accordance with the requirements of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification (see 6.7).

3.4 Design and construction. The configuration and physical dimensions of the switch shall be as specified (see 3.1). The switch shall consist of an assembly with one or more optical input and optical output ports. Switches shall be constructed so that an externally exerted force (either mechanical or elec-

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trical) alters the optical signal path as specified. External port construction shall consist of either connectors or pigtails as the interface to the fiber optic system. Switches with the same Part or Identifying Number (PIN) shall be physically and functionally interchangeable.

3.4.1 Fiber optic pigtail. When specified, switches shall be supplied with fiber or cable pigtails as specified (see 3.1). One end of the pigtail shall be permanently mounted in the switch housing, and the mounting construction may provide strain relief for the pigtail in order to meet the requirements of 3.5.2.1 or 3.5.2.2. The pigtail length shall be not less than 1 meter. No metal components shall be used within the pigtail, except for mounting to the switch housing or termination connector.

3.4.2 Fiber optic connectors. The fiber optic connector type shall be as specified (see 3.1).

3.4.3 Actuator and actuation method. The actuator and the actuation method (latching or nonlatching) shall be as specified (see 3.1).

3.4.4 Operating hardware. Any required operating hardware, such as knobs or levers, shall be as specified (see 3.1).

3.4.5 Electrical connections. Connections, such as screws or solder, shall be as specified (see 3.1).

3.4.5.1 Pin terminals.

3.4.5.1.1 Terminal strength. When tested in accordance with 4.6.2.1.1.1, terminals shall reveal no evidence of breakage, loosening, or relative motion between the electrical terminals and the body of the switch.

3.4.5.1.2 Solderability. When tested in accordance with 4.6.2.1.1.2, terminal solderability shall meet the temperature and time requirements as specified in MIL-STD-883, method 2003, or as specified (see 3.1).

3.4.5.2 Other terminals.

3.4.5.2.1 Terminal strength (see 4.6.2.1.2.1). When tested in accordance with 4.6.2.1.2.1, terminals shall reveal no evidence of breakage, loosening, or relative motion between the electrical terminals and the body of the switch.

3.4.5.2.2 Solderability (see 4.6.2.1.2.2). When tested in accordance with 4.6.2.1.2.2, terminal solderability shall meet the requirements of MIL-STD-454.

3.4.6 Mounting hardware. Each switch shall be provided with mounting hardware if specified (see 3.1).

3.4.7 Seals. Grommets, O-rings, interfacial seals, boots, gaskets, or any other combination of sealing devices, as needed by the particular switch design, shall be provided as specified (see 3.1).

3.4.8 Enclosures. As specified (see 3.1), the type of enclosure shall be either watertight or hermetic (see 3.5.3.5 and 3.5.3.6).

3.4.9 Finish. When tested in accordance with 4.6.1.1, the switch finish shall show no peeling or chipping of the plating or finishes, nicks, burrs, or other surface blemishes.

3.4.10 Dimensions. When tested in accordance with 4.6.1.1, the dimensions of the switch shall be as specified (see 3.1).

3.4.11 Weight. When tested in accordance with 4.6.2.2, the weight of the switch shall be as specified (see 3.1).

3.4.12 Color. When tested in accordance with 4.6.1.1, the color of the switch shall be as specified (see 3.1).

3.4.13 Maintainability. These switches are regarded as lowest replaceable units and, as such, are not repairable nor shall they require maintenance.

3.4.14 Electrical pigtails. When specified, switches shall be supplied with electrical power pigtails in accordance with MIL-W-16878. One end of the pigtail shall be permanently mounted in the switch housing, and the mounting construction may provide strain relief for the pigtail in order to meet the requirements of 3.5.2.1. The pigtail length shall be not less than 1 meter. Polyvinyl Chloride (PVC) shall not be used for any part of the pigtail.

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3.5 Performance requirements. The performance requirements shall be defined in terms of optical, mechanical, environmental, electrical, and chemical properties.

3.5.1 Optical properties.

3.5.1.1 Insertion loss. When tested in accordance with 4.6.3.1, the insertion loss (see 6.6.4) between each input port and each possible associated output port shall be not greater than the limit specified (see 3.1).

3.5.1.2 Change in optical transmittance. When tested in accordance with 4.6.3.2, the change in optical transmittance shall be not greater than 0.5 decibel (dB) for any specified mechanical or environmental requirement. The periodicity of the measurement shall be appropriate for the test mentioned and as approved by the qualifying activity.

3.5.1.3 Return loss. When tested in accordance with 4.6.3.3, the return loss (see 6.6.11) shall be not greater than minus 30 dB.

3.5.1.4 Crosstalk. When tested in accordance with 4.6.3.4, for switches with two or more optical paths (or channels), the sum of signal power levels of passive channels shall be below the level of the active channel by at least 40 dB.

3.5.1.5 Ambient light susceptibility (see 6.6.3). When tested in accordance with 4.6.3.5, the ambient light optical power from any one port or sum of ports shall be not greater than minus 50 dBm.

3.5.1.6 Switching characteristics.

3.5.1.6.1 Operating time. When tested in accordance with 4.6.3.6.1, the operating time (see 6.6.7) shall be as specified (see 3.1).

3.5.1.6.2 Operating bounce time. When tested in accordance with 4.6.3.6.2, the operating bounce time (see 6.6.8) shall be as specified (see 3.1).

3.5.1.6.3 Release time. When tested in accordance with 4.6.3.6.3, the release time (see 6.6.9) shall be as specified (see 3.1).

3.5.1.6.4 Release bounce time. When tested in accordance with 4.6.3.6.4, the release bounce time (see 6.6.10) shall be as specified (see 3.1).

3.5.1.6.5 Repeatability. When tested in accordance with 4.6.3.6.5, and unless otherwise specified (see 3.1), the change in optical transmittance shall not deviate more than 0.25 dB from the initial value.

3.5.1.6.6 Switch stability. When tested in accordance with 4.6.3.6.6 and unless otherwise specified (see 3.1), the change in optical transmittance for each possible path through the switch shall not deviate more than 0.25 dB from the initial value.

3.5.2 Mechanical properties. The following mechanical properties shall be met without physical damage to the switch.

3.5.2.1 Cable pigtails. Unless otherwise specified (see 3.1), switches with optical fiber or electrical cable pigtails shall meet the applicable mechanical properties specified in 3.5.2.1.1 through 3.5.2.1.5.

3.5.2.1.1 Impact (fiber optic and electrical cable pigtails). When tested in accordance with 4.6.4.1.1, the switch shall not be visibly damaged or otherwise rendered unfit for operational use. The change in optical transmittance shall not exceed the value specified in 3.5.1.2 after the test.

3.5.2.1.2 Twist (fiber optic pigtails only). When tested in accordance with 4.6.4.1.2, cable-to-switch seals shall not be rendered inoperable nor shall any other physical damage occur. The change in optical transmittance shall not exceed the value specified in 3.5.1.2 during and after the test.

3.5.2.1.3 Cable seal flexing (fiber optic and electrical cable pigtails). When tested in accordance with 4.6.4.1.3, cable-to-switch strain relief mechanisms shall prevent loss of environmental sealing material or other damage which may impair the switch operation.

3.5.2.1.4 Crush resistance (fiber optic pigtails only). When tested in accordance with 4.6.4.1.4, the switch shall show no evidence of broken parts or accessories, electrical failure, or damage to seals. The change in optical transmittance shall not exceed the value specified in 3.5.1.2 during and after the test.

3.5.2.1.5 Cable pull-out force. The switches shall meet the requirements of 3.5.2.1.5.1 and 3.5.2.1.5.2 as applicable.

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c weight percent finely divided metallic silver.
a maximum of

3.5.2.1.5.1 Fiber optic pigtail pull-out. When tested in accordance with 4.6.4.1.5, the minimum cable-to-switch pull-out strength shall be 100 newtons. There shall be no evidence of cable jacket damage, cable clamp failure, cable-to-switch seal damage, distortion or bending of switch parts, or cable disengagement from the clamp. The change in optical transmittance shall not exceed the value specified in 3.5.1.2 during and after the test.

3.5.2.1.5.2 Electrical pigtail pull-out. When tested in accordance with 4.6.4.1.5, the minimum cable-to-switch pull-out strength shall be 50 Newtons. There shall be no evidence of cable jacket damage, cable clamp failure, cable-to-switch seal damage, distortion or bending of switch parts, or cable disengagement from the clamp. The switch shall meet the requirements of 3.5.4 during and after the test.

3.5.2.2 Connectorized. Unless otherwise specified (see 3.1), switches with fiber optic connectors as the port construction shall meet the mechanical properties specified in 3.5.2.2.1 through 3.5.2.2.3.

3.5.2.2.1 Impact. When tested in accordance with 4.6.4.2.1, the switch shall not be visibly damaged or otherwise rendered unfit for operational use. The change in optical transmittance shall not exceed the value specified in 3.5.1.2 after the test.

3.5.2.2.2 Crush resistance. When tested in accordance with 4.6.4.2.2, the switch shall show no evidence of broken parts or damage to seals. The change in optical transmittance shall not exceed the value specified in 3.5.1.2 during and after the test.

3.5.2.2.3 Connector pull-out force. When tested in accordance with 4.6.4.2.3, pull-out strength shall be 100 Newtons. There shall be no evidence of connector-to-switch seal damage, distortion or bending of switch parts, or connector disengagement from the switch. The change in optical transmittance shall not exceed the value specified in 3.5.1.2 after the test.

3.5.2.3 Operational life. When tested in accordance with 4.6.4.2.4, the operational life of the switch shall be defined as 200,000 operations (50,000 for manually operated switches), where each operation is a complete switching cycle. A complete switching cycle shall be considered to include all possible combinations. The change in optical transmittance shall be less than 0.2 dB after the test.

3.5.3 Environmental properties. The switch shall meet all requirements as specified herein. The operating temperature range and storage temperature range shall be as shown in table I and as specified (see 3.1).

3.5.3.1 Temperature life. When tested in accordance with 4.6.5.1, the switch shall show no evidence of dimensional change, opening of seals, cracking or crazing of components, or any form of physical damage which will impair performance. The change in optical transmittance shall not exceed the value specified in 3.5.1.2 after the test.

TABLE I. Temperature ranges.

Range	Operating (°C)	Storage (°C)
1	-28 to +65	-40 to +70
2	-28 to +85	-40 to +85

3.5.3.2 Thermal shock. When tested in accordance with 4.6.5.2, the switch shall be fully functional and show no evidence of physical degradation. The temperature extremes shall be the specified storage temperature extremes (see 3.1). The change in optical transmittance shall not exceed the value specified in 3.5.1.2 after the test.

3.5.3.3 Temperature-humidity cycling. When tested in accordance with 4.6.5.3, the switch shall show no evidence of deterioration of parts or materials, loosening of finishes, physical distortion, corrosion of metals, entrapment of moisture, separation of bonded surfaces, or other physical damage. The change in optical transmittance shall not exceed the value specified in 3.5.1.2 during and after the test.

3.5.3.4 Salt spray (corrosion). When tested in accordance with 4.6.5.4, the switch shall show no evidence of salt penetration into the switch seals nor evidence of corrosive effects.

3.5.3.5 Water pressure (for watertight enclosures only). When tested in accordance with 4.6.5.5, the switch shall reveal no evidence of the penetration of water into the sealed region of the switch. The insertion loss shall be not greater than 0.5 dB in excess of the initial measured value (see 3.5.1.1).

3.5.3.6 Hermetic (for hermetic enclosures only). When tested in accordance with 4.6.5.6 and when a hermetic enclosure is specified, the switch shall be permanently sealed to prevent transmission of air, gas, or moisture.

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3.5.3.7 Fluid immersion. When tested in accordance with 4.6.5.7, the switch shall reveal no evidence of swelling or softening of material, no loss of sealing capability, nor any other effects detrimental to the operation of the switch.

3.5.3.8 Sand and dust. When tested in accordance with 4.6.5.8, the switch shall show no evidence of physical degradation.

3.5.3.9 Shock. When tested in accordance with 4.6.5.9, the switch shall show no evidence of physical damage that impairs its specified function. The forces exerted on the switch shall not cause the switch to inadvertently throw, latch, or unlatch. The change in optical transmittance shall not exceed the value specified in 3.5.1.2 during and after the test. Discontinuities shall not have a magnitude greater than 0.5 dB and a duration greater than 50 μs.

3.5.3.10 Vibration. When tested in accordance with 4.6.5.10, the switch shall show no evidence of physical damage that impairs its specified function. The forces exerted on the switch shall not cause the switch to inadvertently throw, latch, or unlatch. The change in optical transmittance shall not exceed the value specified in 3.5.1.2 during and after the test. Discontinuities shall not have a magnitude greater than 0.5 dB and a duration greater than 50 μs.

3.5.3.11 Nuclear radiation resistance (switches using lenses only). When tested in accordance with 4.6.5.11, the switch shall meet the requirements of 3.5.1.2.

3.5.3.12 Temperature cycling. When tested in accordance with 4.6.5.13, the switches shall show no evidence of mechanical damage, loosening of component parts, separation of bonded surfaces, or other damage and the optical performance shall be within the limits specified in 3.5.1.1 and 3.5.1.2. The change in optical transmittance shall be less than 0.25 dB.

3.5.3.13 Electromagnetic effects. When tested in accordance with 4.6.5.12, the switch shall meet the radiated emission requirements, RE102, except that the field strength shall be not greater than 1 V/m, and the radiated susceptibility requirements, RS103, of MIL-STD-461, part 5, as specified (see 3.1). The application shall be as specified (see 3.1). The change in optical transmittance shall be less than 0.25 dB during the test.

3.5.4 Electrical properties. If applicable to the switch design, electrical properties shall be as specified (see 3.1).

3.5.4.1 Electrical input signals. The switch shall meet all specified requirements when using electrical input signals as specified (see 3.1).

3.5.4.1.1 Voltage and current levels. When tested in accordance with 4.6.6.1, the voltage and current levels shall be as specified (see 3.1).

3.5.5 Chemical properties.

3.5.5.1 Fungus resistance. When tested in accordance with 4.6.7.1, the switch shall be in accordance with MIL-STD-454, requirement 4, for fungus-inert materials. Switches that are not in accordance with requirement 4 of MIL-STD-454 shall meet grade I classification of MIL-STD-810, method 508.

3.5.5.2 Flame survivability. When tested in accordance with 4.6.7.2, the switch shall not produce explosive fumes or cause any dripping material to ignite.

3.6 Transportability. The switch operational performance specified herein shall not be affected following prolonged storage or shipment under environmental conditions for shipment and storage specified herein, and as required for packaging for shipment (see 5.1).

3.7 Identification marking (see 4.6.1.1). Unless otherwise specified (see 3.1), the switch shall be identified by a marking applied to the outer surface of the switch. The identification marking shall consist of the following:

a. Specification sheet PIN.
b. Manufacturer's code designation.
c. The words "Fiber Optic Switch".
d. Date code (four-digit - week, year).
e. Port identification (input and output ports).

Identification marking shall be in accordance with MIL-STD-1285. Unless otherwise specified (see 3.1), the identification marking characters shall be not less than 2 millimeters (mm) in height. The identification marking shall be permanent and legible in accordance with MIL-STD-1285. Identification marking shall be applied with the vertical axes of the printed characters perpendicular to the longest dimension of the switch housing. Characters shall remain legible for the intended lifetime of the switch. Unless otherwise

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specified (see 3.1), the switch accessory parts shall be identified by a legible and permanent marking applied in accordance with MIL-STD-1285.

3.8 Workmanship. When tested in accordance with 4.6.1.1, the switch shall be free of manufacturing flaws that would degrade performance, inhibit proper connection to interfacing elements, and otherwise yield an inferior product. The following are minimal considerations and are not intended to exclude other pertinent workmanship requirements:

a. Outer surface shall be free of rust, burnt areas, abrasions, holes, roughened areas, bulges, thin spots, and indentations.

b. Inner parts shall be free of fractures, holes, discontinuities, bulges, and thin spots.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-STD-45662.

4.1.3 Reliability assurance program. A reliability assurance program shall be established and maintained in accordance with MIL-STD-790. Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualifications inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.2.1 Toxicological data and formulation. The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.3 Inspection conditions. Unless otherwise specified (see 3.1), all inspections shall be performed in accordance with the test conditions specified in applicable portions of EIA/TIA-455 and as specified herein.

4.4 Qualification inspection. Qualification inspection shall consist of the examinations and tests in table II and as specified (see 3.2 and 6.4).

4.4.1 Sample. A switch sample shall be submitted for each switch construction for which qualification approval is desired. The sample submitted shall be eleven switches of the same PIN, unless otherwise specified (see 6.2).

4.4.2 Inspection routine. The sample shall be subjected to the inspections specified in table II in the order shown. All sample units shall be subjected to the inspection of group I. The sample shall then be divided into the three remaining groups. The sample units shall then be subjected to the applicable inspections for their particular group. After completion of sample testing, all units from non-destructive testing shall be resubjected to group I testing. Any switch failing any inspection shall not be subjected to further inspection.

4.4.3 Failures. One or more failures shall be sufficient cause for refusal to grant qualification approval.

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4.4.4 Retention of qualifications. To retain qualification, the contractor shall forward a report at 12 and 36-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

a. A summary of the results of the tests performed for inspection of product for delivery (group A and group B) indicating, as a minimum, the number of lots that have passed and the number that have failed. The results of tests of all reworked lots shall be identified and accounted for. Summary shall be submitted at 12 month intervals.

b. A summary of the results of tests performed for periodic inspection (group C), including the number and mode of failures. The summary shall include results of all periodic inspection tests performed and completed during the 36-month period. If the test results summary indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

4.4.5 Failure to submit a report. Failure to submit the report within 60 days after the end of each 12 and 36-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the contractor shall immediately notify its qualifying activity at any time during the 12 and 36-month period that the inspection data indicate failure of the qualified product to conform to the requirements of the specification.

4.4.6 No production. In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods there has been no production, the manufacturer will be required, at the discretion of the qualifying activity, to submit this qualified product for testing in accordance with the initial qualification inspection requirements of 4.5.1.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified for group A inspection (table III), group B inspection (table IV), and group C inspection (table V) (see 6.4 and 6.5).

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspection.

TABLE II. Qualifications inspection.

Examination or test	Requirement paragraph	Test method paragraph	Sample
Group I			
Visual and mechanical examination	3.4, 3.4.1 to 3.4.5, 3.7, 3.8	4.6.1.1	11
Finish	3.4.9	4.6.1.1	1/
Dimensions	3.4.10	4.6.1.1	1/
Weight	3.4.11	4.6.2.2	1/
Color	3.4.12	4.6.1.1	1/
Identification marking	3.7	4.6.1.1	1/
Insertion loss	3.5.1.1	4.6.3.1	1/
Return loss	3.5.1.3	4.6.3.3	1/
Crosstalk	3.5.1.4	4.6.3.4	1/
Ambient light susceptibility	3.5.1.5	4.6.3.5	1/
Switching characteristics	3.5.1.6	4.6.3.6	1/
Voltage and current	3.5.4.1.1	4.6.6.1	1/
Group II			
Electrical connections	3.4.5	4.6.2.1	1
Electromagnetic effects	3.5.3.13	4.6.5.12	2/
Fungus resistance	3.5.5.1	4.6.7.1	2/
Flame survivability	3.5.5.2	4.6.7.2	1

See footnotes at end of table.

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TABLE II. Qualification inspection - Continued.

Examination or test	Requirement paragraph	Test method paragraph	Sample
Group III			
Temperature life	3.5.3.1	4.6.5.1	1
Thermal shock	3.5.3.2	4.6.5.2	3/
Temperature cycling	3.5.3.12	4.6.5.13	3/
Temperature-humidity cycling	3.5.3.3	4.6.5.3	3/
Salt spray	3.5.3.4	4.6.5.4	3/
Fluid immersion	3.5.3.7	4.6.5.7	3/
Sand and dust	3.5.3.8	4.6.5.8	2
Vibration	3.5.3.10	4.6.5.10	4/
Shock	3.5.3.9	4.6.5.9	4/
Water pressure	3.5.3.5	4.6.5.5	2/
Hermetic	3.5.3.6	4.6.5.6	4/
Nuclear radiation resistance	3.5.3.11	4.6.5.11	3
Group IV			
Switches (cable pigtails)	3.5.2.1.1	4.6.4.1.1	2
Impact	3.5.2.1.2	4.6.4.1.2	5/
Twist	3.5.2.1.3	4.6.4.1.3	5/
Cable seal flexing	3.5.2.1.4	4.6.4.1.4	5/
Crush resistance	3.5.2.1.5	4.6.4.1.5	5/
Cable pull-out force			
Group V	3.5.2.2.1	4.6.4.2.1	2
Switches (connectorized)	3.5.2.2.2	4.6.4.2.2	5/
Impact	3.5.2.2.3	4.6.4.2.3	5/
Crush resistance			
Connector pull-out force	3.5.2.3	4.6.2.4	5/
Group VI (all switches)			
Operational life			

1/ The same samples shall be used as in the visual and mechanical inspection.
2/ The same sample shall be used as in the electrical connections inspection.
3/ The same sample shall be used as in the temperature life inspection.
4/ The same sample shall be used as in the sand and dust inspection.
5/ The same sample shall be used as in the respective impact inspection.

4.5.2 Group A inspection. Group A inspections shall follow the order shown in table III.

4.5.2.1 Sampling plan. Group A inspection shall be performed on 100 percent of delivered product. There shall be no failures.

4.5.2.2 Disposition of sample units. Sample units which have failed any of the group A inspection tests shall not be delivered on any order.

4.5.2.3 Rejected sample units. If a sample unit is rejected under the finish, color, or identification marking inspections, the manufacturer may rework it to correct the defects and resubmit for reinspection.

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TABLE III. Group A inspection.

Examination or test	Requirement paragraph	Test method paragraph
Visual and mechanical examination	3.4, 3.4.1 to 3.4.5, 3.7, 3.8	4.6.1.1
Finish	3.4.9	4.6.1.1
Dimensions	3.4.10	4.6.1.1
Weight	3.4.11	4.6.2.2
Color	3.4.12	4.6.1.1
Identification marking	3.7	4.6.1.1
Insertion loss	3.5.1.1	4.6.3.1
Return loss	3.5.1.3	4.6.3.3
Crosstalk	3.5.1.4	4.6.3.4
Switching characteristics	3.5.1.6	4.6.3.6
Voltage and current	3.5.4.1.1	4.6.6.1

4.5.3 Group B inspection. Group B inspection shall consist of the inspections specified in table IV. Group B inspections shall be made on units that have passed the group A inspection.

TABLE IV. Group B inspection.

Inspection	Requirement	Test method
Thermal shock	3.5.3.2	4.6.5.2
Switches (cable pigtails)		
Twist	3.5.2.1.2	4.6.4.1.2
Cable seal flexing	3.5.2.1.3	4.6.4.1.3
Cable pull-out force	3.5.2.1.5	4.6.4.1.5
Switches (connectorized)		
Connector pull-out force	3.5.2.2.3	4.6.4.2.3

4.5.3.1 Sampling plan. Sampling shall be in accordance with MIL-STD-105, special inspection level S-4. There shall be no failures.

4.5.3.2 Rejected lots. If an inspection lot is rejected, the manufacturer may rework it to correct the defects, or screen out the defective units and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.3.3 Disposition of sample units. Sample units which have passed all group B inspection may be delivered on the contract or the purchase order, if the lot is accepted. Any connector or connector part deformed or otherwise damaged during testing shall not be delivered on the contract or purchase order.

4.5.4 Periodic inspection. Periodic inspection shall consist of group C inspection. The periodic inspections shall be used for qualification verification, and except where the results of the inspections show noncompliance with the applicable requirements (see 4.5.5.4), delivery of inspection lots which have passed group B inspection shall not be delayed pending the results of group C inspection. Group C inspection shall be performed every 36 months following notification of qualification acceptance.

4.5.5 Group C inspections. Samples submitted to group C inspections shall consist of the inspections in table V. Samples submitted to group C inspection shall have satisfactorily completed all group A and group B inspections. After completion of group C inspections, the samples shall be resubjected to group A inspection.

4.5.5.1 Sampling plan. Group C inspection shall be performed on switches of the same PIN 3-year after initial qualifications and within each 3-year period thereafter. Eight sample units shall be selected. The sample units shall be representative of production over the entire 3-year period and shall be selected from

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more than one production lot.

4.5.5.2 Failures. If one or more sample units fail to pass group C inspection, the sample lot shall be considered to have failed.

4.5.5.3 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract or purchase order.

TABLE V. Group C inspection.

Inspection	Requirement	Test method
Group I		
Electrical connections	3.4.5	4.6.2.1
Electromagnetic effects	3.5.3.13	4.6.5.12
Flame survivability	3.5.5.2	4.6.7.2
Group II		
Temperature life	3.5.3.1	4.6.5.1
Temperature cycling	3.5.3.12	4.6.5.13
Temperature-humidity cycling	3.5.3.3	4.6.5.3
Salt spray	3.5.3.4	4.6.5.4
Fluid immersion	3.5.3.7	4.6.5.7
Vibration	3.5.3.10	4.6.5.10
Shock	3.5.3.9	4.6.5.9
Water pressure	3.5.3.5	4.6.5.5
Hermetic	3.5.3.6	4.6.5.6
Group III		
Impact	3.5.2.1.1	4.6.4.1.1
Crush resistance	3.5.2.1.4	4.6.4.1.4
Operational life (all switches)	3.5.2.3	4.6.2.4

4.5.5.4 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, on all units of product which can be corrected and which are manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Groups A and B inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

4.6 Methods of inspection.

4.6.1 General testing practice.

4.6.1.1 Visual and mechanical examination. Unless otherwise specified (see 3.1), visual and mechanical examinations shall be performed in accordance with EIA 455-13 to verify that the design, construction, physical characteristics, dimensions, marking, and workmanship are in accordance with the applicable requirements.

4.6.1.2 Optical test methods. The use of optical test methods to assess the influence of test exposures on performance of switches requires that they be connected to lengths of fiber. Unless otherwise specified (see 3.1), the optical properties inspections shall be performed at a wavelength of 1300 ±20 nanometers (nm). The fiber should be cabled, that is, the switches should be attached to lengths of cable of the type specified in MIL-C-85045. The cable lengths shall be not greater than 30 meters. The optical fiber components exposed at the test set (source, detector, or coupler) may be connectorized for access to the test set. For multimode fiber, the source used shall be noncoherent. Light launch conditions shall be in accordance with table VI.

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TABLE VI. Light launch conditions.

Fiber type	Launch conditions
Single mode	30-mm diameter mandrel
Multimode	70/70 restricted or equivalent and overfill (see 6.6.12)

Any optical power detection method may be used if the method is sufficiently sensitive to measure the differential optical power levels as specified (see 3.5.1.1), and if the method provides repeatable readings (less than 3 percent variation).

4.6.1.3 Switch position. The switch position, latched or unlatched, for the environmental and mechanical tests shall be as specified (see 3.1).

4.6.1.4 Equivalent test methods. The use of equivalent test methods is allowed subject to the following conditions:

a. The allowance of an equivalent method is specified in this specification.

b. The manufacturer has conducted both test methods and has submitted complete test data to the preparing activity.

c. The preparing activity has approved the use of that method by that manufacturer.

4.6.2 Construction inspection.

4.6.2.1 Electrical connections.

4.6.2.1.1 Pin terminals.

4.6.2.1.1.1 Terminal strength (see 3.4.5.1.1). Electrical terminals shall be tested and accepted in accordance with MIL-STD-883, method 2004. Test condition A shall be used for tension; condition B1 shall be used for stress; and condition B2 shall be used for fatigue.

4.6.2.1.1.2 Solderability (see 3.4.5.1.2). The electrical leads of the switch shall be tested in accordance with MIL-STD-883, method 2003, to verify solderability to circuit boards.

4.6.2.1.2 Other terminals.

4.6.2.1.2.1 Terminal strength (see 3.4.5.2.1). The terminals shall be tested and accepted in accordance with MIL-STD-202, method 211. Test condition A and test condition E shall be used. The load for test condition A shall be 22.25 newtons.

4.6.2.1.2.2 Solderability (see 3.4.5.2.2). The terminals shall be tested and accepted in accordance with MIL-STD-202, method 208.

4.6.2.2 Weight (see 3.4.11). The switch weight shall be verified using scales of sufficient range and accuracy appropriate for the specified weight and tolerance.

4.6.3 Optical properties inspection.

4.6.3.1 Insertion loss (see 3.5.1.1). The insertion loss shall be verified in accordance with EIA-455-34, method C using overfill conditions for both SM and MM and 70/70 restricted conditions or equivalent for MM only. A 30 mm diameter mandrel shall be used for SM.

4.6.3.2 Change in optical transmittance (see 3.5.1.2). The change in optical transmittance shall be verified in accordance with EIA 455-20 or equivalent (see 4.6.1.4). Light launch conditions shall be 70/70 or equivalent for MM and a 30 mm diameter mandrel for SM.

4.6.3.3 Return loss (see 3.5.1.3). The return loss shall be verified in accordance with EIA-455-107.

4.6.3.4 Crosstalk (see 3.5.1.4). The switch crosstalk shall be verified in accordance with EIA 455-42. The test temperature shall be +25°C.

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4.6.3.5 Ambient light susceptibility (see 3.5.1.5). The ambient light susceptibility shall be verified in accordance with EIA 455-22, with the exception that the output power in the "on" state shall be referenced to 1 milliwatt. The test temperature shall be +25°C.

4.6.3.6 Switching characteristics (see 3.5.1.6).

4.6.3.6.1 Operating time (see 3.5.1.6.1). The switch shall be actuated with an input mechanical force, input electrical signal, or other latching or nonlatching force of step amplitude equal to the lower, center, and higher specified input mechanical force, electrical voltage, or other force. The operating time shall be verified by analyzing the optical output (see figure 1).

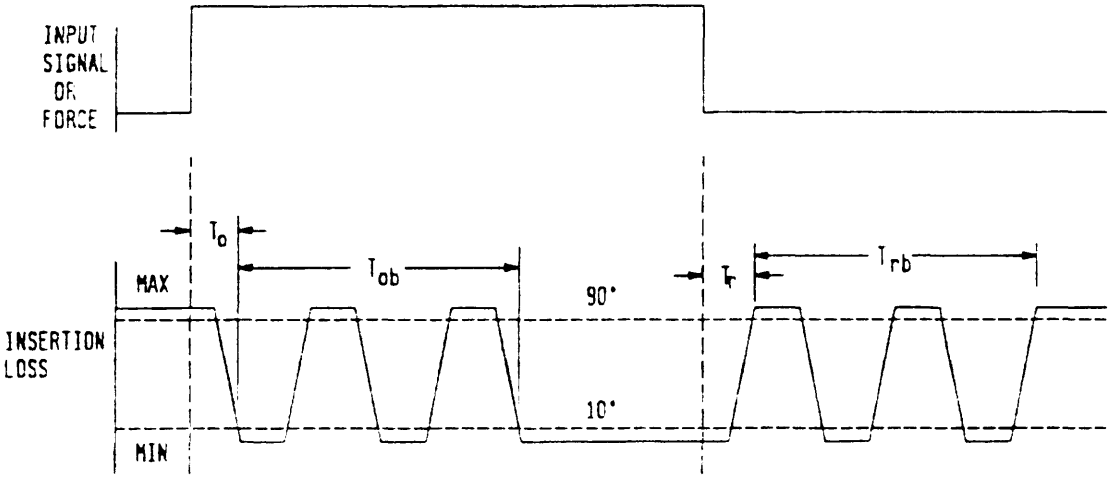
4.6.3.6.2 Operating bounce time (see 3.5.1.6.2). The switch shall be actuated with an input mechanical force, input electrical signal, or other latching or nonlatching force of step amplitude equal to the lower, center, and higher specified input mechanical force, electrical voltage, or other force. The operating bounce time shall be verified by analyzing the optical output (see figure 1).

4.6.3.6.3 Release time (see 3.5.1.6.3). An already actuated switch shall be used for this test. The actuator force shall be removed so that the removal is of step amplitude equal to the opposite of the lower, center, and higher specified input mechanical force, electrical force, or other force. The release time shall be verified by analyzing the optical output (see figure 1).

4.6.3.6.4 Release bounce time (see 3.5.1.6.4). An already actuated switch shall be used for this test. The actuator force shall be removed so that the removal is of step amplitude equal to the opposite of the lower, center, and higher specified input mechanical force, electrical force, or other force. The release bounce time shall be verified by analyzing the optical output (see figure 1)

4.6.3.6.5 Repeatability (see 3.5.1.6.5). The switch shall be operated for 500 cycles under ambient conditions where each cycle shall consist of all possible input port to output port configurations. The change in optical transmittance shall be monitored during the test (see 4.6.3.2).

4.6.3.6.6 Switch stability (see 3.5.1.6.6). The switch shall be monitored for a period of 1 hour while in the activated position. The change in optical transmittance shall be verified during and after the test (see 4.6.3.2).



T_o = Operating time
T_{ob} = Operating bounce time
T_r = Release time
T_{rb} = Release bounce time

FIGURE 1. Switching characteristics.

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4.6.4 Mechanical properties inspection.

4.6.4.1 Cable pigtails.

4.6.4.1.1 Impact (see 3.5.2.1.1). Switches shall be tested as follows:

a. The impact test facility shall consist of a clamp for securing cable pigtails and a steel block, .500 inch minimum thickness, as shown on figure 2.

b. The impact test shall be conducted as follows: The cable pigtail is clamped so that, with the switch hanging under its own weight, the end of the switch extends to the center of the steel block. The distance between the bottom of the cable clamp and the bottom of the hanging switch shall be greater than or equal to 1 meter. The switch is then raised to the height of the clamp, and with the cable extended, released so as to strike the block. The switch shall be dropped 8 times.

c. The change in optical transmittance shall be measured after the test.

4.6.4.1.2 Twist (see 3.5.2.1.2). The switch shall be tested in accordance with EIA 455-36, with the exception that the test sample shall be a properly pigtailed switch. The number of cycles shall be 50. The tensile load shall be 5 newtons. The change in optical transmittance shall be verified during and after the test (see 4.6.3.2). At the completion of the test, the switch shall be visually examined (see 4.6.1.1). This test shall be performed for each cable pigtail.

4.6.4.1.3 Cable seal flexing (see 3.5.2.1.3). The switch shall be tested and accepted in accordance with MIL-STD-1344, method 2017, except the test sample shall be a pigtailed switch. The number of cycles shall be 50. At the completion of the test, the switch shall be visually examined (see 4.6.1.1).

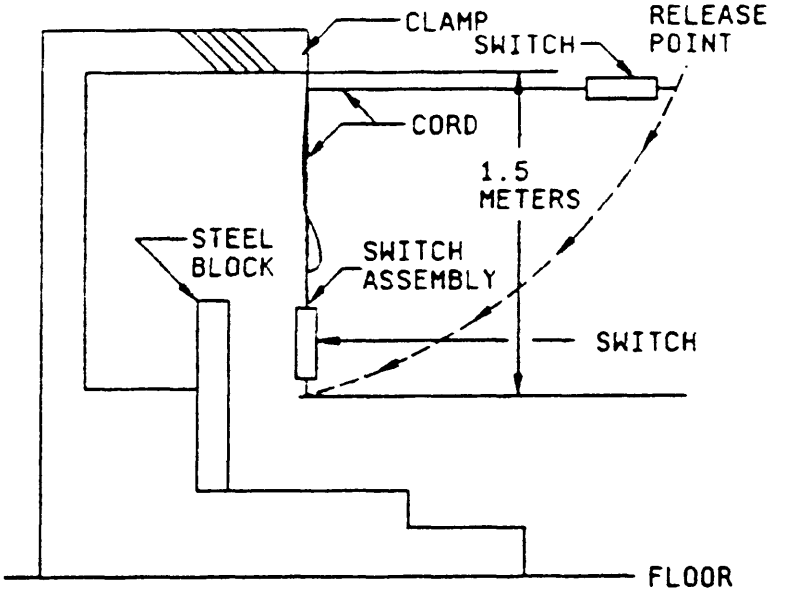


FIGURE 2. Impact test facility.

4.6.4.1.4 Crush resistance (see 3.5.2.1.4). The switch shall be tested in accordance with MIL-STD-1344, method 2008, with the exception that the test sample shall be a properly pigtailed switch. The test load shall be 500 newtons, and the number of loads shall be one. The change in optical transmittance shall be verified during and after the test (see 4.6.3.2). At the completion of the test, the switch shall be visually examined (see 4.6.1.1).

4.6.4.1.5 Cable pull-out force (see 3.5.2.1.5). The switch shall be tested in accordance with MIL-STD-1344, method 2009 with the exception that the test sample shall be a switch. The axial tensile load shall be applied and shall be maintained for 10 minutes. The change in optical transmittance shall be

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verified during and after the test (see 4.6.3.2). At the completion of the test, the switch shall be visually examined (see 4.6.1.1).

4.6.4.2 Connectorized.

4.6.4.2.1 Impact (see 3.5.2.2.1). Switches shall be tested as follows:

a. The impact test facility shall consist of a clamp for securing cable pigtails and a steel block, .500 inch minimum thickness, as shown on figure 2.

b. The impact test shall be conducted as follows: Jumper cables mated to the switch are clamped so that, with the switch hanging under its own weight, the end of the switch extends to the center of the steel block. The distance between the bottom of the cable clamp and the bottom of the hanging switch shall be greater than or equal to 1 meter. The switch is then raised to the height of the clamp, and with the cable extended, released so as to strike the block. The switch shall be dropped eight times.

c. The change in optical transmittance shall be measured after the test.

4.6.4.2.2 Crush resistance (for package S switches only) (see 3.5.2.2.2). The switch shall be tested in accordance with MIL-STD-1344, method 2008, with the exception that the test sample shall be a switch with connectors as the port construction. The test load shall be 500 newtons, and the number of loads shall be one. The change in optical transmittance shall be verified during and after the test (see 4.6.3.2). At the completion of the test, the switch shall be visually examined (see 4.6.1.1).

4.6.4.2.3 Connector pull-out force (see 3.5.2.2.3). The switch shall be tested in accordance with EIA-455-6, with the exception that the test sample shall be a connector on the switch. The axial tensile load shall be applied and shall be maintained for 10 minutes. The change in optical transmittance shall be verified after the test (see 4.6.3.2). At the completion of the test, the switch shall be visually examined (see 4.6.1.1).

4.6.4.2.4 Operational life (see 3.5.2.3). The switch shall be subjected to 200,000 cycles of operation (50,000 for manually operated switches). Switching rate shall be as specified (see 3.1). The change in optical transmittance shall be verified during and after the test (see 4.6.3.2).

4.6.5 Environmental properties inspection.

4.6.5.1 Temperature life (see 3.5.3.1). The switch shall be tested in accordance with MIL-STD-202, method 108. The exposure time shall be 240 hours. The exposure temperature shall be +110°C. The change in optical transmittance shall be verified after the test (see 4.6.3.2). At the completion of the test, the switch shall be visually examined (see 4.6.1.1).

4.6.5.2 Thermal shock (see 3.5.3.2). The switch shall be tested for five cycles in accordance with DOD-STD-1678, method 4020. The high and low temperatures shall correspond to the specified storage temperature extremes (see 3.1). The change in optical transmittance shall be verified after the test (see 4.6.3.2). At the completion of the test, the switch shall be visually examined (see 4.6.1.1).

4.6.5.3 Temperature-humidity cycling (see 3.5.3.3). The switch shall be tested in accordance with DOD-STD-1678, method 4030. The subcycle shall be included in the test. The change in optical transmittance shall be verified during and after the test (see 4.6.3.2). At the completion of the test, the switch shall be visually examined (see 4.6.1.1).

4.6.5.4 Salt spray (corrosion) (see 3.5.3.4). The switch shall be tested and accepted in accordance with MIL-STD-1344, method 1001. The exposure time shall be 96 hours and the exposure temperature shall be +35°C. At the completion of the test, the switch shall be visually examined (see 4.6.1.1).

4.6.5.5 Water pressure (see 3.5.3.5). The switch shall be immersed in water at any applied pressure of 0.1 megapascals (equivalent to a depth of 10.4 meters) for 48 hours at a temperature of +10°C to +35°C. The insertion loss shall be verified after the test (see 4.6.3.1). At the completion of the test, the switch shall be visually examined (see 4.6.1.1).

4.6.5.6 Hermetic (see 3.5.3.6). The switch shall be tested and accepted for hermetic seal in accordance with MIL-STD-883, method 1014. Test condition A1 shall be used for the seal test, and test condition C shall be used for the gross leak test. Pigtails may be removed prior to the test.

4.6.5.7 Fluid immersion (see 3.5.3.7). The switch shall be tested in accordance with MIL-STD-1344, method 1016. Each switch shall be immersed in each immersion fluid as specified in table VII for a period of 24 hours. Each switch shall be completely dried before further testing. At the completion of the test, the switch shall be visually examined (see 4.6.1.1).

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TABLE VII. Immersion test fluids.

Fluids	Specification	Test temperature (°C)
Fuel oil	MIL-F-16884	33-37
Fuel (JP-5)	MIL-T-5624	20-25
Isopropyl alcohol	TT-I-735	20-25
Hydraulic fluids	MIL-H-5606	48-50
Lubricating oils	MIL-L-17331	73-77
	MIL-L-23699	73-77
Coolant 1/	None	20-25
Sea water	ASTM-D-1141	20-25

1/ Monsanto coolant 25 or equivalent.

4.6.5.8 Sand and dust (see 3.5.3.8). The switch shall be tested in accordance with MIL-STD-202, method 110. The change in optical transmittance shall be verified during and after the test (see 4.6.3.2). At the completion of the test, the switch shall be visually examined (see 4.6.1.1).

4.6.5.9 Shock (see 3.5.3.9). The switch shall be tested in accordance with MIL-S-901, grade A, type A class I. Printed circuit board mounted switches shall be tested in accordance with MIL-STD-883, method 2002, test condition B. Unless otherwise specified (see 3.1), the signal discontinuity shall be monitored in accordance with EIA-455-32 during the test, with equipment having a time resolution sufficient to resolve discontinuities of duration not less than 50 μs (see 4.6.3.2). At the completion of the test, the switch shall be visually examined (see 4.6.1.1). The change in optical transmittance shall be verified after the test.

4.6.5.10 Vibration (see 3.5.3.10). The switch shall be tested in accordance with MIL-STD-1344, method 2005, test conditions II and VI. For test condition VI, test condition C shall apply and the duration shall be 30 minutes per axis. Test condition I shall be used during the low frequency portion of test condition II. Unless otherwise specified (see 3.1), the signal discontinuity shall be monitored in accordance with EIA-455-32 during the test, with equipment having a time resolution sufficient to resolve discontinuities of duration not less than 50 μs (see 4.6.3.2). At the completion of the test, the switch shall be visually examined (see 4.6.1.1). The change in optical transmittance shall be verified after the test.

4.6.5.11 Nuclear radiation resistance (switches with internal lenses only)(see 3.5.3.11). The switch shall be tested in accordance with the appendix to this specification. Unless otherwise specified (see 3.1), tests shall be performed at a wavelength of 1300 ±20 nm, at the low and high operating temperatures (see 3.5.3), and at +20°C. The total dose shall be as specified (see 6.2). The change in optical transmittance shall be verified during and after the test (see 4.6.3.2).

4.6.5.12 Electromagnetic effects (see 3.5.3.13). The switch shall be tested for effects of electromagnetic emissions in accordance with MIL-STD-462 as applicable and as specified (see 3.1). The change in optical transmittance shall be verified during and after the test (see 4.6.3.2).

4.6.5.13 Temperature cycling (see 3.5.3.12). The cable shall be tested in accordance with the steps listed in table VIII. The change in optical transmittance shall be monitored continuously during and after the test.

4.6.6 Electrical properties inspection.

4.6.6.1 Voltage and current levels (see 3.5.4.1.1). The voltage and current levels shall be tested and accepted as specified (see 3.1).

4.6.7 Chemical properties inspection.

4.6.7.1 Fungus resistance (see 3.5.5.1). Switches that do not meet the requirements of fungus-inert materials in accordance with MIL-STD-454, requirement 4, shall be tested and accepted for exposure to fungus in accordance with MIL-STD-810, method 508.

4.6.7.2 Flame survivability (see 3.5.5.2). The switch shall be tested in accordance with MIL-STD-1344, method 1012, test condition A, with the exception that the test sample shall be a switch instead of a connector. The change in optical transmittance shall be verified after the test (see 4.6.3.2).

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TABLE VIII. Temperature cycling steps.

Step	Temperature (°C)	Duration (hours)
1	Room ambient	24
2	Ramp to the low operating temperature	2
3	Low operating temperature	8
4	Ramp to 25 ±2	2
5	25 ±2	6
6	Ramp to high operating temperature	1
7	High operating temperature	6
8	Ramp to 25 ±2	1
9	25 ±2	6
10	Repeat steps 2 through 9 four times a for total of five cycles	

4.6.8 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.8.)

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-28786. The level of packaging shall be as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The switches covered by this specification are intended for use in any application where their performance characteristics are required. The switches are for installation in shipboard systems within the limitations of applicable performance requirements.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of the specification.
(b) Switch classification (see 1.2).
(c) Quantity of switches required.
(d) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(e) Applicable specification sheet number, title, and date.
(f) Equivalent test methods, if other than as specified (see 4.6.1.4).
(g) Total dose radiation required (see 4.6.5.11).
(h) Level of packaging required (see 5.1).
(i) Applicable specification sheet PIN (see 6.9).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID number	DID title	Suggested tailoring
4.4	DI-T-4902	Qualification inspection report	----
4.5	DI-T-5329	Inspection and test reports	----

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

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6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in the applicable Qualified Products List QPL-24725, whether or not such products have actually been so listed by that date. The attention of contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 55Z3, Washington, DC 20362-5101, however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center, ATTN: DESC-EQ, Dayton, OH 45444-5000.

6.4.1 Lot size. Lot size may be specified contractually if desired.

6.5 Quality conformance. Quality conformance inspections require contractual definition of the overall test program, including sample sizes and lot sizes, if appropriate (see 4.5).

6.6 Definitions.

6.6.1 Actuation method. Actuation method is the way in which a motive force must be applied to a switch in order to place it into its various states.

6.6.2 Actuator. An actuator is a device that provides the motive force that must be applied to a switch to place it into its various states.

6.6.3 Ambient light susceptibility. Ambient light susceptibility is the optical power (dBm) that enters the switch from ambient illumination incident upon the switch.

6.6.4 Insertion loss. Insertion loss is the radiant power loss (dB) caused by absorption, scattering, diffusion, leaky waves, dispersion, microbends, macrobends, reflection, radiation, or other causes when a switch is inserted into the system.

6.6.5 Latching. Latching is a switch actuation method that requires a specific signal to place the switch in a position where it remains until another actuating force or signal is applied.

6.6.6 Nonlatching. Nonlatching is a switch actuation method that requires continuous application of an activating force or signal to cause the switch to remain in the position that it is in.

6.6.7 Operating time. Operating time is the time between the instant an applied actuating force or signal reaches a specified value and the instant the transmittance changes 90 percent of its maximum value.

6.6.8 Operating bounce time. Operating bounce time is the time between the instant an applied actuating force or signal reaches a specified value and the instant the transmittance permanently changes 90 percent of its maximum value minus the operating time.

6.6.9 Release time. Release time is the time between the instant an applied actuating force or signal ceases and the instant the transmittance has changed by 90 percent of its maximum value.

6.6.10 Release bounce time. Release bounce time is the time interval between the cessation of the applied actuating force or signal and the time that the optical transmittance has permanently changed by 90 percent of its maximum value, minus the release time.

6.6.11 Return loss. Return loss is the optical power (dB) that is reflected back toward the source of the optical power by the switch.

6.6.12 70/70 restricted launch. A 70/70 restricted launch is a beam optics launch with a 70 percent spot size and source aperture equal to 70 percent of the fiber numerical aperture.

6.6.13 Switching time. Switching time is the time interval between initiation of the switch actuation method (or initiation of the switching actuation) to the settling of the active output to within 10 percent of the nominal steady-state transmittance. Mathematically, the switching time, T_{sw}, is:

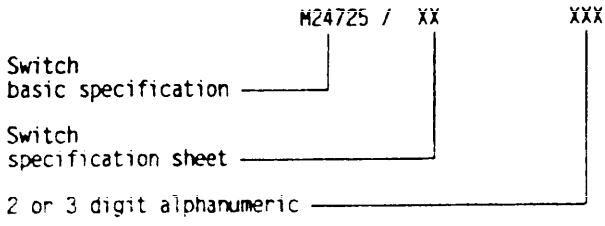
$$T_{sw} = \text{(operating time)} + \text{(operating bounce time)}$$

6.7 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing address for submission of data are listed in FED-STD-313.

6.8 Subcontracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

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6.9 PIN. Switches should contain only the following:



Examples: M24725/1-01
M24725/2-01S

6.10 Part designator. A part designator, if specified (see 3.1), should include classifications (see 1.2) as follows:

1. Type (see 1.2.1).
2. Style (see 1.2.2).

Example: 4C.

6.11 Subject term (key word) listing.

Actuation method
Actuator
Bypass switch
Change in optical transmittance
Exchange switch
Fiber optic communications
Fiber optic component
Insertion loss
Latching
Nonlatching
On/off switch
Operating time
Operating bounce time
Release time
Release bounce time
Repeatability
Selector switch
Switch stability

6.12 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes

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APPENDIX

PROCEDURE TO MEASURE NUCLEAR RADIATION EFFECTS IN FIBER OPTIC COMPONENTS

10. SCOPE

10.1 Scope. This procedure describes a method for the determination of transient and steady-state nuclear radiation effects in optical switches, cables or devices. It should be noted that either or both of the measurements can be performed depending on the data that is desired. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. TEST EQUIPMENT

30.1 Test equipment. The following test equipment shall be used as required for testing in the configuration as shown. Optical, mechanical and thermal stability of the test setup is necessary to facilitate movement of the test setup enabling the test sample to be subjected to environmental tests following the coupling loss test.

30.1.1 Radiation source. Three types of sources depending on the type of measurement being performed are as follows.

30.1.1.1 Transient radiation pulse response. A source shall produce at least 450 rads (Si) total dose with a pulse width of 1 μs or less. (A bremsstrahling x-ray beam may be obtained by placing a tantalum target in front of the electron beam.) An example of the test equipment is shown on figure 3.

30.1.1.2 Steady-state response. The container is placed in a Cobalt-60 source to deliver gamma rays at doses from 3000 ±100 rads (Si) minimum to 18,000 ±100 rads (Si) maximum. An example of test equipment is shown on figure 4.

30.1.1.3 Neutron fluence measurements. The container is placed in a reactor which delivers a density of fast neutrons (<1 MeV) at a nominal fluence of 10^{12}n/cm^2/s. An example of the test equipment is shown on figure 5.

30.1.2 Light source. A light source shall be used with an output power of about 100 μW, but not more than 1 mW for multimode fiber. NOTE: If a source of more than 100 μW is used, photo-bleaching phenomenon may occur. Lower power levels may be usable depending on the dynamic range of the overall system. For single mode fiber, the power may have to be increased to get adequate light coupled into the fiber. For multimode fiber, wavelengths of 850 ±10 nm and 1300 ±10 nm shall be used unless otherwise specified. For single mode fiber, wavelengths of 1310 ±10 nm and 1550 ±10 nm shall be used unless otherwise specified.

30.1.3 Input optics. A system of optical components shall be used to create a monochromatic (less than or equal to 10 nm Full Width Half Maximum (FWHM)), substantially constant radiance spot larger in diameter than that of the specimen, unless otherwise specified by another fiber optic test procedure. A means of verifying the alignment of the endface shall be provided, such as a mounted telescope tube. Optical filters may be used to limit the spectral bandwidth of the source. For multimode fibers, an equilibrium mode simulator shall be utilized (see 30.1.7).

30.1.4 Specimen support and alignment. A means of supporting the input end of the specimen shall be arranged to allow stable and repeatable positioning without introducing significant deformation. Suitable means shall be provided to align the input endface with respect to the launch radiation.

30.1.5 Cladding mode stripper. Cladding mode stripper is a device which extracts only cladding modes and is used reasonably near the ends of the fiber under test. In an effort to completely characterize the fiber, two measurements will be performed for each test: One measurement without any cladding mode strippers and one with cladding mode strippers on each end of the test fiber. It is felt for short lengths of fiber there will be cladding mode propagation in actual system use; hence, no mode strippers are used. If an index-matching fluid which is hygroscopic (such as glycerin) is used, it shall be replenished frequently. If the fiber is to be held against a flat, horizontal surface containing the index-matching fluid, small weights may be used to hold it in intimate contact.

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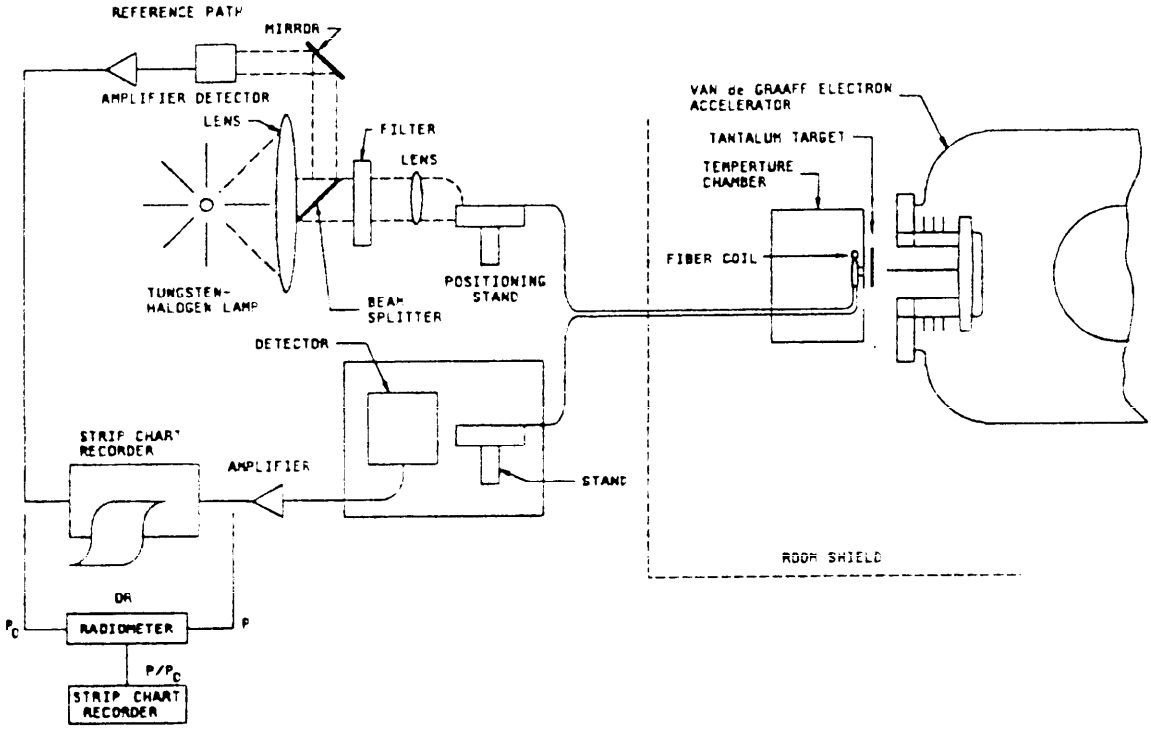


FIGURE 3. Prompt gamma radiation test instrumentation example.

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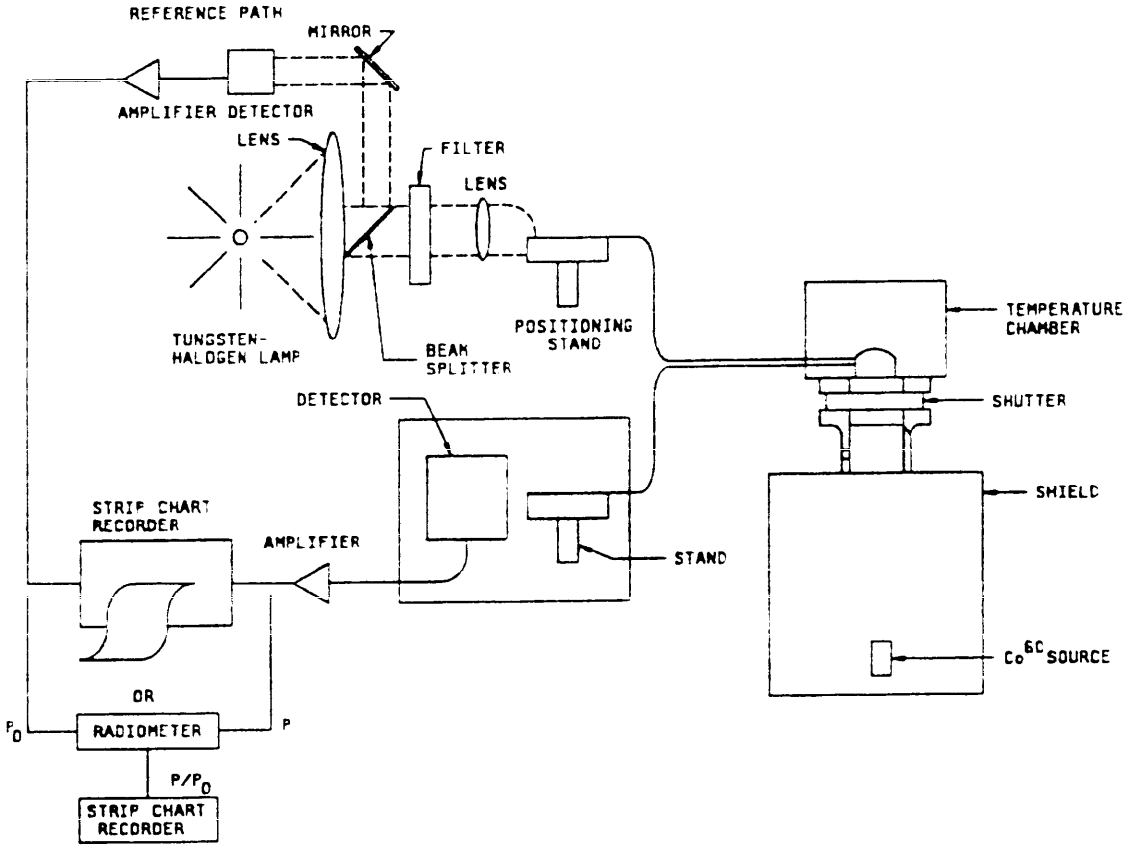


FIGURE 4. Total gamma radiation test instrumentation example.

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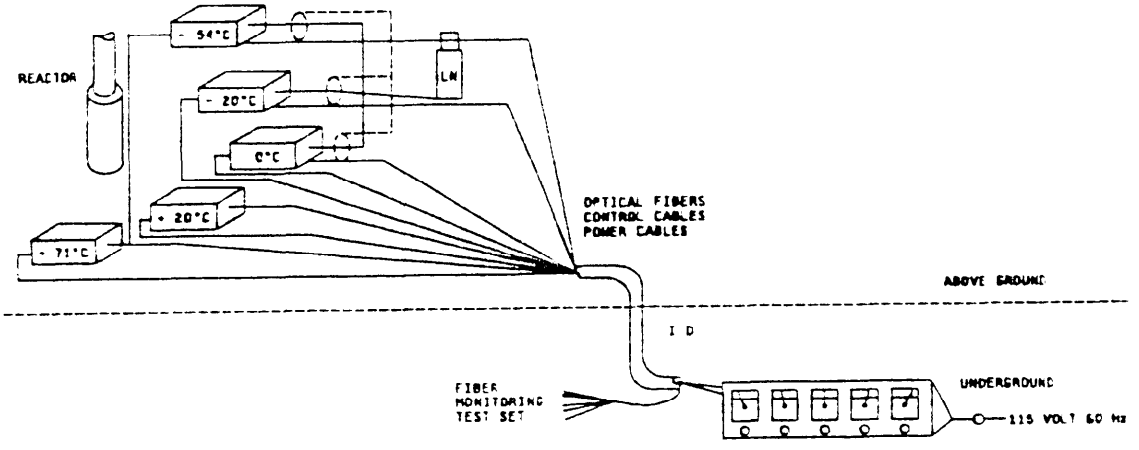


FIGURE 5. Neutron fluence radiation test apparatus example.

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30.1.6 Detector, signal detection electronics. A detector which is linear within 5 percent over the range of detected intensities shall be used. A typical system may include a PIN photodiode, with a rise/fall time of 1 ns or lower; an oscilloscope with a bandwidth of 200 MHz for the transient response measurement, and a chart recorder for the steady-state measurement.

30.1.7 Equilibrium modal simulator (EMS). An EMS is used to attenuate higher order modes and to arrive at a steady-state condition, only for multimode, graded-index fiber. There are a number of methods to accomplish this: A mandrel-wrapped mode filter or a restricted launch condition created by beam optics are examples. Refer to EIA-455-50, "Light Launch Conditions for Long-Length Graded-Index Optical Fiber Spectral Attenuation Measurements", for directions on how to measure for the light launch conditions. In an effort to completely characterize the fiber, an EMS will be used with the cladding mode strippers and not used during the measurement done without cladding mode strippers.

30.1.8 Beam splitter. A beam splitter or fiber optic coupler shall be used to couple the light beam into the specimen fiber and provide a reference signal. The splitter or coupler shall divert a small portion of the input light to a reference detector. The reference path shall be such that the system fluctuations can be monitored during the entire time that measurements are taken (see figure 3).

30.1.9 Fiber support. A means of stable support for the input end of the fiber shall be arranged (for example, a vacuum check can be used). The total length of the fiber shall be handled such that microbending attenuation is kept to a minimum.

30.1.10 Radiation dosimeter. The detection system will be dependent on the type of radiation source used.

30.1.10.1 Transient radiation pulse. A Faraday cup and a plastic scintillator detector shall be used to determine the transient radiation shape. A 200 MHz bandwidth oscilloscope shall be part of the scintillator detection system. Thermoluminescent LiFs or CaFs Crystal Detectors (TLDs) shall be used to measure the total dose received by the specimen fiber.

30.1.10.2 Long duration radiation. TLDs shall be used to measure the total dose received by the specimen fiber.

30.1.11 Collimator. By placing the fiber behind the collimator, the radiation beam will be collimated with respect to the fiber. This will limit the length of fiber subject to radiation.

30.1.12 Interference filter. A 10 nm bandwidth interference filter shall be placed between the output of the fiber and detector. The filter shall have a center wavelength that matches the optical source.

30.1.13 Temperature controlled container. The temperature controlled container shall have the capability of temperatures from -55°C to +125°C. The chamber shall be remotely controlled or programmable to allow change of chamber temperature. The chamber shall be capable of a rate of change of 2°C minimum. The chamber shall be able to maintain a temperature +10°C ±1°C.

40. TEST SAMPLE

40.1 Specimen. The specimen to be measured shall be a component such as a switch or a length of representative fiber or cable. The minimum length of the specimen outside the chamber shall be that which is required to install the fiber from the source to the detector, unless otherwise specified (see 3.1). The maximum length shall be 10 times the exposed length or 1 km, whichever is shorter, unless otherwise specified (see 3.1).

40.2 Reference path. The reference path shall be through the beam splitter and associated optics (see figures 3, 4, or 5).

40.3 Measured test length. The length of the test fiber shall be recorded in the results, if different from the 1 km length specified above.

40.4 Sample configuration. Unless otherwise specified, the specimen for the steady-state evaluation shall be spooled on a reel with a 4 to 12-inch drum diameter, which will depend upon the sample length. Allowance shall be made for the unspooling of a measured length of fiber from each end of the reel to allow for attachment to the optical power measurement equipment.

40.5 Exposed length. The length of the exposed section of specimen will vary for the test setup used, the nature of the specimen, etc. For a particular measurement system, for example, the experimenter may find that up to 250 m of exposed specimen may be necessary to obtain an accurate loss measurement. In any event, the length of specimen exposed shall be noted in the test report.

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40.5.1 Exposed length (minimum) (see 3.1). Unless otherwise specified, the recommended minimum exposure lengths given below shall be used for measurement as noted:

Measurement	Specimen type	Exposure section length
Transient	Glass-on-glass	50 ±5 m
	Plastic or plastic-clad silica	50 ±5 m
Steady-state	Glass-on-glass	50 ±5 m
	Plastic or plastic-clad silica	50 ±5 m

50. TEST PROCEDURE

50.1 High dose calibration. If the specimen is to be tested at a high dose-rate, calibration of the radiation source shall be done before the specimen is setup in the system. Four TLDs shall be placed in the area of exposure and the center of the TLDs shall be placed where the axis of the specimen will be placed. (Four are used to get a representative average value.) A dose equal to the actual test dose shall be used to calibrate the system.

50.2 Calibration accuracy. To maintain the highest possible accuracy in measuring the test dose, the TLDs shall not be used more than once.

50.3 Transient radiation pulse calibration. If a transient radiation pulse is to be used, the pulse width and shape shall be calibrated to obtain the required radiation exposure.

50.4 Numerical aperture (NA). Determine the far-field NA of the specimen under test by the method of FOTP-47, Output Far-Field Radiation Pattern Measurements. This is done so that the input optics can be adjusted to obtain an overfilled NA and core conditions.

50.5 System exposure effects on calibration. Before the specimen is put into the radiation exposure system, but while the specimen is connected to the optical measurement system, the radiation source shall be run to determine its effect on the system. If the measurement is affected by operation of the radiation source, the measurement system shall be shielded against the interference and recalibrated to assure the proper operation. NOTE: The measurement system should be outside the radiation chamber.

50.6 Specimen preparation. If the input to the specimen or the specimen itself is a fiber, than the fiber shall be prepared so that the endfaces are smooth and perpendicular to the fiber axis, in accordance with EIA/TIA-455-57, "Optical Fiber End Preparation and Examination".

50.7 Specimen mounting facilities. The specimen shall be placed in the test setup (see figures 3, 4, or 5) such that the test section shall be exposed to the beam.

50.7.1 Fiber alignment. For fiber input, the input end of the fibers shall be placed in a positioning device and aligned. The output end shall be positioned so that all light exiting the fiber impinges on the detector.

50.8 Sample precondition. The sample shall be preconditioned for 2 hours prior to testing at the test temperature.

50.9 Coupled power. The coupled power shall be determined at the test wavelengths by adding the power loss of the fiber or fiber optic device, or both, to the absolute power at the detector-end of the long length.

50.9.1 Attenuation measurement. Perform an attenuation measurement on the test sample in accordance with the methods of EIA/TIA-455-46, "Spectral Attenuation Measurement, for long-length, graded-index optical fibers". For graded index fibers, or in accordance with EIA-455-20, "Measurement of Change in Optical Transmittance", for other fibers. Record the attenuation, A_L of the sample.

50.9.2 System restoration. Restore the system as outlined in 50.7.1 to get ready for the actual measurement.

50.10 System alignment, radiation source off. The input end of the specimen under test shall be positioned so maximum optical power is received at the detector. Once set, the input launch conditions should not be changed during the balance of the test.

50.11 Preirradiation power measurement. Prior to irradiation, measure the output power at all test wavelengths at the test temperature that will be used.

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50.12 Irradiation exposure. Once the system has been aligned, the irradiation sequence may begin (either steps 50.13, 50.14, or 50.15).

50.13 Transient radiation pulse measurements. The test transient radiation effects are determined by subjecting test specimen to at least 450 rads (Si) total dose in a time period of 1 μs or less. During the radiation pulse, the detector may be inoperative to eliminate the effects of the luminescence caused by the radiation. Begin optical power measurements 100 ns after the radiation pulse is terminated. The measurement schedule thereafter shall be as follows:

Time after radiation terminated		Measurement frequency
From	To	Every
100 ns	1 μs	100 ns
1 μs	10 μs	1 μs
10 μs	100 μs	10 μs
100 μs	10 μs	100 μs
10 ms	1 s	10 ms
1 s	10 s	100 ms
10 s	10 minutes	10 s
10 minutes	30 minutes	30 s

50.14 Steady-state measurements. Steady-state nuclear effects shall be determined by subjecting the test specimen to dose rates of 50 rads (Si) per second with a total dose of up to 10,000 rads (Si). A 60°C source or equivalent shall be used to provide the radiation. During the radiation exposure, the power out measurement is recorded continuously. A chart recorder shall be connected to the detection system so that a continuous power measurement is made. The chart recorder shall be setup such that the detection signal does not exceed the limits of the recorder.

50.15 Neutron fluence measurements. Neutron fluence effects shall be determined by subjecting the test specimen to a neutron exposure at a nominal fluence of 10^{12} n/cm^2/s in a pulse with a duration ≤ 1 ms. The intent of this test is to measure the effect of neutron radiation. The effect of gamma radiation shall be minimized. During the neutron pulse, the detector may be inoperative to eliminate the effects of the luminescence caused by the radiation.

50.16 Relative detected power measurement. For transient and neutron testing, measurements shall continue until the output power is within 5 percent of the preirradiated level or to the point where the output power value is constant for three consecutive measurements. For the steady-state testing, measurements shall continue until the output power is within 5 percent of the preirradiated level or to the point where there is no perceivable change in output power. Repeat steps 50.6 through 50.12 for the required test temperatures. It will be necessary to use a new nonirradiated specimen for each temperature required.

50.17 Radiation hardening. If radiation hardening is required by the specification sheet (see 3.1), the following operational sequence shall be followed:

a. Steps 50.1 through 50.12 shall be performed.

b. The specimen shall be subjected to radiation two orders of magnitude greater than the level at which the specimens will be tested.

c. The specimen shall be tested after the output power has decreased to within 50 percent of the preirradiated level or to the point where there is no perceivable change in output power.

d. Steps 50.12 through 50.17 shall be performed to determine the radiation hardening characteristics of the specimen.

60. CALCULATIONS

60.1 Far-field numerical aperture calculations. See EIA-455-47.

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60.2 Coupled power into fiber. To find the coupled power into the specimen from the source, the following formula is used:

$P_C = (P_O) / 10 (A_b/10)$

where P_O = Power output from long length.

A_b = Attenuation of specimen, as determined 50.9.1.

60.3 Change in optical transmittance calculations. The change in optical transmittance (A) shall be calculated for each wavelength by using the following formula:

$A \text{ (in dB)} = -10 \log 10 (P_A/P_B)$

where P_A is the measured output of the specimen fiber after irradiation

P_B is the measured output of the specimen fiber before irradiation

In case of the radiation hardening section, P_B is the measurement before the last irradiation.

60.4 Normalized change in optical transmittance calculations. The results of the reference signal shall be used to normalized the test results.

$A_{ref} \text{ (in dB)} = -10 \log 10 (P_A1/P_B1)$

where P_A1 is the power at the reference detector at the end of the measurement.

The test results that are normalized for system fluctuations are found by use of the following formula:

$A_{nor} \text{ (in dB)} = A - A_{ref}$

where A_{nor} is the output power of the specimen fiber when normalized against system fluctuations.

70. DOCUMENTATION

70.1 Test report. The following information shall be reported:

a. Operator.

b. Date.

c. Identification of specimen. Fiber characteristics such as fiber composition, fiber dimensions and delta index change shall be reported.

d. Optical source type, wavelength and spectral width.

e. Optical source characteristics, i.e., spot size and beam numerical aperture.

f. Method of stripping cladding modes.

g. Length of specimen of fiber.

h. Length or area of specimen exposed to the radiation.

i. For transient pulse measurement, the behavior of the detector to a 300 picosecond rise time pulse at 850 and 1300 nm shall be documented.

j. The specimen's and reference's output power with respect to time shall be recorded.

k. Characteristics of the interference filters.

l. The calculated input coupled power into the test fiber shall be recorded.

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APPENDIX

80. SUMMARY

80.1 Required test condition. The following shall be specified in the detail specification:

a. Type of specimen material to be tested.

b. Procedure number to be referenced (FOTP).

c. Test conditions.

(1) Temperature.

(2) Length of specimen section exposed.

(3) Type of radiation.

(4) Exposure rate and duration.

(5) Type of dosimeters used.

(6) Type of measurement (transient or steady-state).

(7) Whether radiation hardening is to be tested.

d. Acceptance criteria.

(1) Specimen response characteristics.

(2) Recovery time of specimen.

(3) Allowable permanent damage.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85
NASA - NA

Review activities:
Army - MI, AR
Navy - AS
Air Force - 17
DLA - ES

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 6021-0001)

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-24725A	2. DOCUMENT DATE (YYMMDD) 29 NOVEMBER 1994

3. DOCUMENT TITLE
SWITCHES, FIBER OPTIC, SHIPBOARD, (METRIC), GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER	
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY	
a. NAME COMMANDER NAVAL SEA SYSTEMS COMMAND (SEA 03R42)	b. TELEPHONE (Include Area Code) (1) Commercial (703) 602-0347 (2) AUTOVON DSN 332-0347
c. ADDRESS (Include Zip Code) 2531 JEFFERSON DAVIS HWY ARLINGTON, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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